KEVIN KWAN

Software Engineer II | Conversational AI & Voice Automation
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SUMMARY

Software Engineer II at GEICO — conversational AI and voice automation. Build and operate the Sierra AI billing voice agent and Amazon Connect IVR on AWS, automating routine billing self-service in place of human agents: 436,000+ conversations at ~51% containment, ~30,000 calls/week.

WORK EXPERIENCE

GEICO July 2024 – Present
Atlanta, GA (Remote)
Software Engineer II — Conversational AI & IVR February 2026 – Present
Software Engineer I — Technology Development Program July 2024 – February 2026

- Spearheaded the Sierra AI billing voice agent (live since May 2026) as its largest contributor, containing ~51% of 436,000+ English and Spanish conversations without live-agent transfer (May–Oct 2026) and improving containment from 48% to 52% as volume scaled ~10x to ~30,000 calls/week across payments, rescheduling, AutoPay, and billing questions.
- Delivered end-to-end make-payment flows, processing 22,600+ payments at 92–95% authorization success across saved methods, scheduled payments, policies pending cancellation, and declined- or expired-card recovery; shipped net-new Pause AutoPay, 82,800 reschedules, and 192,000 informational answers at 98% tool success.
- Re-architected the billing agent from regex- and classifier-based routing to fact-based journeys — decomposing every intent into goal-based journeys for distinct self-service needs (~20,000 net lines removed) — and built simulation-based regression suites and automated code-review quality gates.
- Pioneered conversation patterns — locale switching, operator speedbumps/recovery, and gated, goal-based journeys — adopted by the main menu and authentication teams and adapted for the billing customer assistant live chat agent, a separate channel from voice; also standardized platform analytics tagging.
- Resolved 86 of 91 production and pre-release billing bugs at the root cause; reviewed code for 12+ engineers and onboarded two engineers onto the Sierra platform.
- Owned production reliability across release validation, Cyara automated IVR regression testing, outage rollbacks, and postmortems, and drove a full-volume conversation-experience review to resolve top customer-experience gaps.
- Migrated GEICO's legacy Cisco UCCE and Nuance IVR to Amazon Connect on AWS, engineering Java traffic-redirection logic for gradual live-caller cutover, legacy endpoint migration, and Amazon Lex pattern migration.
- Led the Proof of Coverage IVR (~80% containment) and its disaster-recovery rebuild pipeline, and built the AWS self-service payment IVR (~60% containment) with a zero-amount-due flow, automated declined-payment retries, and new card tokenization.
- Engineered asynchronous voice-workflow orchestration using AWS Lambda, API Gateway, DynamoDB, and Amazon Lex (JavaScript code hooks) across Amazon Connect contact flows, with AES-256 encryption for customer PII and payment APIs and Terraform-automated infrastructure deployments.

Publicis Sapient (Publicis Groupe) June 2023 – August 2023
Software Engineer Intern | Atlanta, GA

- Built a WhatsApp AI chatbot for the Travel & Hospitality industry using Google Vertex AI's PaLM LLM, LangChain, MongoDB, and external APIs to deliver live data and recommendations for Atlanta attractions, hotels, and restaurants, achieving 91% response accuracy.
- Developed Python/Flask RESTful APIs bridging MongoDB, the AI chatbot backend, and an interactive map website, automating CI/CD pipelines via GitLab and Docker to deploy on Google App Engine and Firebase Hosting.

Georgia Tech EcoCAR January 2022 – May 2023
Software Engineer, Connected & Automated Vehicles (V2X Subteam) | EcoCAR (Electric Vehicle) Mobility Challenge | Atlanta, GA

- Designed a camera/sensor state machine for automated stop/go decisions across traffic-light scenarios, helping Georgia Tech secure 1st place in the U.S. Department of Energy's 2022 EcoCAR Mobility Challenge.

PROJECTS & OPEN SOURCE

- **PreMiD (Open Source):** Authored the Stake.us activity (TypeScript) for PreMiD, a browser-extension activity tracker displaying live website usage publicly through Discord status — merged April 2024, follow-up enhancement Nov. 2025.
- **Unity3D Fishing Rod Motion:** Casting and line-reeling physics in Unity (C#): a Verlet rope model rendered via Line Renderer — a modular, extensible base environment for other developers' physics sims and games; 49 stars on GitHub.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C#, C++
Conversational AI & Voice: Sierra AI, Amazon Connect, Amazon Lex, LangChain, Google Vertex AI, OpenAI, Cyara, LLM Evaluation, Agentic Workflows, Tool Calling, State Machines
AWS & Cloud Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, CloudWatch), Serverless, Event-Driven Systems, Asynchronous Orchestration, Disaster Recovery, Google Cloud Platform (App Engine, Firebase)
Web, APIs & Data: REST APIs, Flask, Node.js, React, Next.js, Tailwind CSS, MySQL, MongoDB, Cloud Firestore
DevOps & Tooling: Terraform, Docker, CI/CD, Git, GitHub, GitLab, SonarQube, Cursor, Claude Code

EDUCATION

Georgia Institute of Technology — Atlanta, GA August 2021 – May 2024

- Bachelor of Science in Computer Science, College of Computing; Threads: Intelligence and Media; *summa cum laude* GPA: 3.9
- Zell Miller Scholarship, Graduated with Highest Honors, Dean's List, Faculty Honors